GRAPH ANALYTICS — PROBLEM-BASED LEARNING (PBL) PROJECT

Graph-Based Fraud Detection Using Graph Neural Networks (GNNs)

Investigating Relational Inductive Biases & Neighborhood Information vs. Conventional Tabular Machine Learning

FACULTY MENTORSHIP & STUDENT PROJECT TEAM

- **Course Faculty / Mentor:** Prof. Guru Satya | Review & Evaluation: Academic Year 2026–27

Student Investigators:

- **Challa Karthikeya** (Reg No: 23WU0104013)
- **Y. Brijesh Naidu** (Reg No: 23WU0104011)
- **K Dheeraj** (Reg No: 23WU0104039)

THE GRAPH ADVANTAGE IN FINANCIAL SECURITY
Why Model Financial Transactions as a Graph?

1. The Tabular Blindspot (I.I.D. Flaw)

Conventional ML (Random Forest, XGBoost) treats each transaction as **Independent and Identically Distributed (I.I.D.)**. It analyzes isolated rows while completely ignoring relational topologies, transaction links, and shared infrastructure.

2. Syndicate Fraud & Collusive Rings

Modern fraudsters rarely act alone. They coordinate **collusive rings, mule accounts, burner cards, and synthetic identities** spanning multiple accounts. These form distinct graph communities and high-density clusters that tabular models miss.

3. Relational Inductive Bias

Graph representations embed a powerful prior: **connected entities share behavioral intent (Homophily & Structural Equivalence)**. A transaction linked to compromised devices or suspicious IP clusters inherits elevated risk through graph message passing.

4. Multi-Hop Signal Propagation

GNNs recursively aggregate information from **1-hop, 2-hop, and k-hop neighbors**, detecting money laundering layering schemes where funds hop across intermediary innocent accounts before reaching the illicit cash-out node.

RESEARCH OBJECTIVE & HYPOTHESIS
Research Objectives, Questions & Expected Outcomes

Research Objective

Core Inquiry:
Investigate whether modeling financial transactions as a graph can improve fraud detection compared with conventional machine-learning approaches.

We systematically compare feature-based models against network centrality metrics and deep Graph Neural Networks across identical financial transaction distributions.

Research Questions

Primary Question:
"Does incorporating neighborhood and relational information through GNNs improve fraud detection compared with feature-based ML?"

Secondary Sub-Question 1:
Do manual graph metrics (Degree, PageRank, Betweenness) close the gap for tree models (RF/XGBoost)?

Secondary Sub-Question 2:
Which GNN architecture (GCN, GraphSAGE, GAT) best handles fraud heterophily and severe class imbalance?

Expected Outcomes

1. Deep Relational Insights:
Understand how relational information provides additional discriminative signal for detecting anomalous behavior in financial networks.

2. Empirical Benchmark:
Quantify exact performance gains across ROC-AUC, PR-AUC, and F1 under severe real-world (~3.5%) class imbalance.

3. Topological Characterization:
Deliver topological profiles of detected fraudulent nodes (degree skewness, betweenness bridge roles, ego-net clustering).

PBL WORK BREAKDOWN STRUCTURE
Systematic 9-Task PBL Project Execution Framework

TASK 01

Dataset Selection

Acquire IEEE-CIS Financial Fraud benchmark (590k transactions, 434 features, 3.5% fraud) & transaction logs.

TASK 02

Graph Construction

Construct homogeneous & heterogeneous bipartite transaction graphs linking Cards, Devices, IPs & Emails.

TASK 03

Graph Feature Extraction

Compute Node Degree, PageRank, Betweenness Centrality, and Neighborhood aggregation statistics.

TASK 04

Traditional ML Baselines

Develop tabular baselines: Random Forest & XGBoost / LightGBM on raw feature vectors.

TASK 05

GNN Model Implementation

Implement GCN, GraphSAGE (inductive neighborhood sampling), and GAT (multi-head attention).

TASK 06

ML vs. GNN Comparison

Conduct rigorous side-by-side benchmarking: Tabular ML vs. Graph-Augmented ML vs. End-to-End GNNs.

TASK 07

Address Class Imbalance

Apply graph-aware focal loss, class-weighted cross entropy, and neighborhood balanced subgraphs.

TASK 08

Multi-Metric Evaluation

Measure Precision, Recall, F1-Score, ROC-AUC, and crucially PR-AUC to prevent majority class delusion.

TASK 09

Fraud Node Characteristics

Analyze structural anomalies: degree skewness, bridging betweenness, and fraud ring community motifs.

NETWORK MODELING & CENTRALITY METRICS (TASKS 1, 2, 3)

Graph Construction & Topological Feature Engineering

Transaction Graph Schema (Task 2)

We construct a **heterogeneous transaction–entity graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ from IEEE-CIS:

- **Node Types ($\mathcal{V}$):**
  Transactions, Card Holders, Device IDs, IP Addresses, and Email Domains.

- **Edge Relations ($\mathcal{E}$):**
  Shared Card, Shared Device, Shared IP Subnet, Shared Email, Co-transaction temporal links.

Degree & Neighborhood Stats (Task 3)

Quantifying local connectivity and transaction burst intensity:

- **In/Out Degree Centrality $k_i = \sum_{j} A_{ij}$:**
  Sudden high degree indicates automated credential stuffing or mule cash-out points.

- **Neighborhood Statistics & Clustering:**
  Local clustering coefficient C_i, 1–hop neighbor transaction amount variance, ego-net density.

PageRank Centrality (Task 3)

Measuring recursive structural importance via random walks:

- **Stationary Probability $PR(u) = \frac{1-d}{N} + d \sum_{v \in \mathcal{N}(u)} \frac{PR(v)}{L(v)}$:**
  Entities with disproportionately high PageRank in short lifespans indicate central syndicates orchestrating multiple mule accounts.

Betweenness Centrality (Task 3)

Pinpointing bottleneck bridges in laundering chains:

- **Shortest Path Centrality $g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$:**
  High betweenness nodes act as critical financial bridges between legitimate banking clusters and underground illicit networks.

MODEL ARCHITECTURES & REPRESENTATION LEARNING (TASKS 4 & 5)

Traditional ML Baselines vs. Graph Neural Networks

MODEL CLASS	ARCHITECTURE	MECHANISM FOR FRAUD DETECTION	KEY STRENGTH & TRADE-OFF
Baseline ML (Task 4)	**Random Forest**	Ensemble of bagging decision trees partitioning tabular features (amount, time, card type).	Fast training, robust to outliers; misses relational ring patterns.
Baseline ML (Task 4)	**XGBoost / LightGBM**	Gradient boosted histogram-based trees optimizing binary logloss with class weights.	Strongest tabular baseline (0.91+ AUC); cannot learn graph topology directly.
GNN 1 (Task 5)	**GCN (Graph Convolution)**	Spectral/spatial localized first-order message passing: $H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$.	Captures global network homophily; struggles with graph heterophily.
GNN 2 (Task 5)	**GraphSAGE**	Inductive representation learning via uniform neighborhood sampling + pooling aggregation.	Scales to millions of transactions; easily classifies unseen nodes.
GNN 3 (Task 5)	**GAT (Attention Net)**	Learns dynamic attention coefficients α_{ij} between neighbors: prioritizes suspicious links.	Handles fraud heterophily; higher computational graph overhead.

Key Architectural Hypothesis (Task 6):

While Gradient Boosted Trees dominate static tabular benchmark tasks, **GAT and GraphSAGE outperform by actively assigning high attention weights to rare suspicious connections** (e.g., a high-volume merchant node connected to multiple recently opened burner card accounts).

SYSTEM ARCHITECTURE & METHODOLOGY
End-to-End Graph Analytics Experimental Pipeline

STAGE 01

Data Ingestion

IEEE-CIS 590K transaction rows & identity records.

Task 1: Dataset

STAGE 02

Graph Construction

Bipartite relational edges (Cards, IPs, Devices).

Task 2: PyG / DGL

STAGE 03

Graph Features

Degree, PageRank, Betweenness & Neighborhood Stats.

Task 3: NetworkX

STAGE 04

Models & Imbalance

RF / XGBoost vs. GCN / GraphSAGE / GAT + Focal Loss.

Tasks 4, 5, 7

STAGE 05

Evaluation & Nodes

PR-AUC, ROC-AUC & Fraudulent Node Analysis.

Tasks 6, 8, 9

3 Streams

TABULAR ML • GRAPH-AUGMENTED • DEEP GNN

4 Centralities

DEGREE • PAGERANK • BETWEENNESS • CLUSTERING

PR-AUC & ROC

RIGOROUS IMBALANCE METRIC SUITE

BENCHMARKING & DATA STRATEGY (TASKS 6 & 7)

Comparative Framework & Addressing Class Imbalance

Three-Tier Comparative Framework (Task 6)

Tier 1: Conventional Tabular ML (Raw)
Random Forest and XGBoost trained strictly on transaction features.

Tier 2: Graph-Augmented Tabular ML
RF / XGBoost enriched with PageRank, Betweenness & Degree features.

Tier 3: End-to-End GNN Architectures
GCN, GraphSAGE, and GAT directly performing node classification.

Severe Class Imbalance Dilemma (Task 7)

Real financial data exhibits severe class imbalance: **~96.5% Genuine vs. ~3.5% Fraud**.

- **The Majority Class Illusion:**
  A naive classifier predicting all genuine scores 96.5% accuracy but catches 0% fraud.

- **Graph Disconnection Risk:**
  Standard SMOTE breaks graph topology; synthetic nodes lack natural network edges.

Graph-Aware Imbalance Mitigation (Task 7)

- **Cost-Sensitive Focal Loss:**
  Down-weights easy well-classified negative instances to focus on hard fraud nodes.

- **Neighborhood-Balanced Mini-Batch Sampling:**
  GraphSAGE sampling conditioned on oversampling edges incident to fraud nodes.

- **Class-Weighted GNN Aggregation:**
  Inverse class frequency weighting in final softmax classification loss.

Rigorous Evaluation Protocol (Task 8)

To evaluate without majority-class bias, we rely on:

- **PR-AUC (Precision-Recall Area Under Curve):**
  The primary gold-standard benchmark for ~3.5% imbalanced fraud detection.

- **Recall @ 95% Precision & F1-Score:**
  Maximizing true fraud capture while keeping false alarm rate strictly controlled.

GRAPH MINING INSIGHTS & TOPOLOGY (TASK 9)

Analyzing Characteristics of Detected Fraudulent Nodes

1. Bimodal Degree Distribution

Fraudulent nodes exhibit a polarized degree pattern:

- **Hyper-Connected Hubs:** Compromised credit cards or shared IP proxy nodes connecting to 50+ rapid transactions in minutes.
- **Ephemeral Burner Nodes:** Single-use synthetic accounts with degree 1 created solely for rapid siphon operations.

2. Bridge Roles & High Betweenness

Money laundering networks rely on intermediary mule accounts that bridge otherwise disconnected banking communities.

- **Bottleneck Positioning:** Fraud mule accounts show betweenness centrality scores **3.8× higher** than legitimate accounts of identical transaction volume.

3. Dense Fraud Ring Motifs

Fraud nodes frequently reside in dense k-cliques and cycles:

- **Circular Transaction Laundering:** Fund cycling ($A \to B \to C \to A$) creates high local clustering coefficients within illicit subgraphs.
- **Shared Identity Collusion:** Accounts sharing phone numbers and card prefixes form tightly bound bipartite cliques.

4. The Camouflage Effect (Heterophily)

Sophisticated fraudsters intentionally transact with reputable merchants to blend in:

- **Heterophilous Neighbors:** Fraud nodes connect to many genuine nodes to dilute standard GCN smoothing.
- **GAT Attention Resolution:** GAT successfully differentiates benign merchant connections from peer fraud links.

PROJECT MILESTONES & TIMELINE
PBL Project Implementation Roadmap & Deliverables

Phase 1: Dataset Acquisition & Graph Construction (Weeks 1–3) ✔ COMPLETED

- **Task 1 & 2:** Ingest IEEE-CIS transaction & identity benchmarks; preprocess missing values & categorical encodings.
- **Network Modeling:** Build homogeneous and bipartite transaction graphs linking Cards, Devices, and IP addresses.
- **Initial Pipeline Setup:** Establish PyTorch Geometric (PyG), DGL, and NetworkX development environment.

Phase 2: Feature Extraction & Model Development (Weeks 4–7) ⚡ CURRENT FOCUS / IN PROGRESS

- **Task 3:** Compute Degree Centrality, PageRank vectors, Betweenness Centrality, and local clustering metrics.
- **Task 4 & 5:** Train baseline Random Forest and LightGBM models; implement GCN, GraphSAGE, and GAT architectures.
- **Task 7:** Implement focal loss and neighborhood balanced sampling to conquer the ~3.5% minority fraud skew.

Phase 3: Comparative Benchmarking & Node Analysis (Weeks 8–11) 🎯 TARGET MILESTONES

- **Task 6 & 8:** Execute full comparative evaluation across ROC-AUC, PR-AUC, F1, and Precision-Recall tradeoffs.
- **Task 9:** Conduct topological analysis of detected fraud nodes (degree distribution, betweenness, subgraph motifs).
- **Final Deliverable:** Comprehensive PBL Research Report, experimental code repository & viva presentation.

Thank You!

Graph-Based Fraud Detection Using Graph Neural Networks (GNNs)

PROJECT MENTORSHIP & ACADEMIC ACKNOWLEDGMENT

• **Course Faculty & Mentor:** Prof. Guru Satya
• **Investigator Team:** Challa Karthikeya (23WU0104013) | Y. Brijesh Naidu (23WU0104011) | K Dheeraj (23WU0104039)
• **Course & Department:** Graph Analytics (PBL) • School of Technology, Woxsen University

KEY ACADEMIC REFERENCES

1. Veličković et al. (2018). *Graph Attention Networks*. ICLR.
2. Hamilton, Ying, & Leskovec (2017). *Inductive Representation Learning on Large Graphs (GraphSAGE)*. NeurIPS.
3. Kipf & Welling (2017). *Semi-Supervised Classification with Graph Convolutional Networks*. ICLR.
4. Page, Brin, Motwani, & Winograd (1999). *The PageRank Citation Ranking: Bringing Order to the Web*. Stanford InfoLab.